Abhilash Shah

Bhopal, MP | +91-9511454951 | abhilashshah2006@gmail.com | LinkedIn | GitHub | Portfolio | LeetCode

EDUCATION

B.Tech in Computer Science and Engineering **Expected 2028**
VIT Bhopal University *Bhopal, Madhya Pradesh*

TECHNICAL SKILLS

Languages:	Python, Java, C++, JavaScript (ES6+), TypeScript, SQL, Bash, HTML, CSS
AI/ML:	Machine Learning, Deep Learning, Recommender Systems, Learning-to-Rank, NLP, Feature Engineering, Model Evaluation, LLMs, Prompt Engineering
ML Libraries:	PyTorch, TensorFlow, XGBoost, scikit-learn, Pandas, NumPy, MLflow
Backend:	FastAPI, Node.js, Express.js, RESTful APIs, WebSockets, JWT, OAuth 2.0, RBAC
Frontend:	React.js, Next.js (App Router), Vite, Tailwind CSS
Databases:	PostgreSQL, Supabase, MongoDB, MySQL, SQLAlchemy (ORM)
Cloud & DevOps:	Docker, CI/CD (GitHub Actions), Vercel, Linux, Git, GitHub
Testing:	pytest, Vitest, Unit and Integration Testing, Postman
Core Concepts:	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, A/B Testing

PROJECTS

VIT Bhopal Sports Facilities — Full-Stack Booking & Access System **Live at VIT Bhopal** | GitHub
Next.js, TypeScript, Node.js, Hono, Supabase (PostgreSQL, RLS, pg_cron), Zod, Tailwind CSS, Vitest, PGlite

- Launched a campus booking platform in **production** serving **1,000+ weekly active users** across **3 sports facilities**, built to **scale** to all **10,000 students**, with a **Next.js** frontend and a **Hono REST API**.
- Eliminated **double bookings** under **concurrent requests** using **partial unique indexes**, per-slot **advisory locks** and **idempotency keys**, so retries on unstable mobile networks never create duplicate bookings.
- Hardened security with **Row-Level Security** on all **16 tables**, **RBAC** for **3 roles**, **Google OAuth/OTP** restricted to college emails, **JWT** verification, **Zod** validation and per-user **rate limiting**.
- Automated **QR check-in**, **waitlist offers** and **no-show release** with idempotent **pg_cron** jobs, plus an **admin analytics dashboard**, all verified by **239 automated tests** on in-memory PostgreSQL.

CartSense — Explainable Product Recommendation Engine GitHub
Python, PyTorch, XGBoost, scikit-learn, MLflow, FastAPI, PostgreSQL, React, TypeScript, Docker, GitHub Actions

- Architected a **two-stage hybrid recommender** on **814K real Amazon reviews** (25K users, 23K products): **4 retrieval models** (ALS, item-kNN, TF-IDF, popularity) generate candidates that an **XGBoost LambdaMART** ranker re-orders using **24 engineered features**.
- Improved **NDCG@10 by 39%** (0.0131 to 0.0182) over the strongest of **6 baselines**, including a **PyTorch Neural Collaborative Filtering** model, on a **leakage-free**, time-based held-out period, with every experiment tracked in **MLflow**.
- Designed an **A/B testing** framework that compares the model with a popularity control using a **two-proportion z-test** on click-through and add-to-cart rates, and solved **cold start** by personalizing for guests after one product view.
- Shipped precomputed, **explainable** recommendations through a versioned **FastAPI REST API** with **8 endpoint groups**, **JWT** and **API-key** authentication, a **retraining pipeline** driven by store events, and **CI/CD** on GitHub Actions.

MeriYatra — Real-Time Public Transit Tracking Platform GitHub
React.js, Vite, Tailwind CSS, Supabase Realtime, PostgreSQL, WebSockets, React-Leaflet, OpenStreetMap, OSRM

- Engineered a **real-time transit tracking** web app that turns any driver's phone into a **live GPS beacon**, letting commuters track **3 types of local transit** (autos, cabs, private buses) with no app install.
- Streamed vehicle locations every **2 seconds** over **WebSockets** with **sub-100 ms latency**, using **broadcast channels** instead of database writes to keep **infrastructure cost near zero**.
- Integrated **OSRM** road routing for live **ETAs** (throttled to one call per **15 seconds**), plus distance-sorted nearby vehicles, shareable tracking links, ride scheduling and a **bilingual (English/Hindi)**, **mobile-first** UI on **Vercel**.
- Implemented **place autocomplete** via **Nominatim geocoding**, a **screen wake lock** for uninterrupted driver broadcasts, **clock-skew-safe** timestamps and automatic **stale-vehicle cleanup** for an accurate live map.

EXPERIENCE

Open Source Contributor (Top 100), GirlScript Summer of Code (GSSoC) **2026**

- Ranked among the **Top 100 contributors** of GSSoC 2026 after merging **150+ pull requests** (features, bug fixes and documentation) across multiple **open-source** repositories.
- Resolved issues across unfamiliar codebases, incorporating maintainer feedback through **Git-based code review** until each pull request met the project's quality standards.

Technical Team Member, 5 College Clubs, VIT Bhopal

- Drove technical execution for **5 college clubs**, delivering **multiple concurrent projects** through strong **collaboration** and ownership.
- Organized technical operations for campus events with **1,000+ participants**, coordinating **cross-functional teams** for smooth execution.

CERTIFICATIONS

- **Oracle Cloud AI Foundations Associate**, Oracle
- **Prompt Engineering**, Infosys Springboard
- **Full Stack Web Development**, Apna College
- **Data Analytics Job Simulation**, Deloitte Australia
- **Introduction to Internet of Things**, NPTEL